Actuation-Aware Sensor Validation for Fault-Resilient Closed-Loop Hydroponic Control

MD. Sameer Iqbal Chowdhury[a], Mikdam-Al-Maad Ronoue[a], Md Asaduzzaman[b], Lafifa Jamal[a,*]

[a] *Department of Robotics and Mechatronics Engineering, University of Dhaka, Dhaka-1000, Bangladesh*
[b] *Department of Agriculture and Food Technology, Kyoto University of Advanced Sciences, Japan*

Abstract

Unlike a conventional detector, AASVR-R can record a suspect measurement while withholding its authority to drive an actuator. This paper proposes AASVR-R, a streaming measurement-to-actuation reliability layer for closed-loop hydroponic control. The method uses robust gates, persistence states, bounded rectification, actuator-response residuals, sensor reliability memory, and risk-aware authorization. Held-out replay on a real NFT hydroponic deployment shows competitive detection, high specificity, and auditable decision traces relative to tuned filtering, process-monitoring, model-based, and one-class baselines. Counterfactual response replay further shows that response residuals can identify no-response and stuck-after-command behavior that single-stream plausibility gates miss. The results identify slow drift and weak response as the principal remaining risks for future deployments with real actuator logs.

Keywords: sensor validation, data rectification, fault detection, process control, hydroponics, actuation

1. Introduction

Hydroponic controlled-environment systems depend on continuous measurement and actuation. In a recirculating Nutrient Film Technique (NFT)

*Corresponding author.

installation, the streams that describe pH, electrical conductivity (EC), air temperature, humidity, water temperature, water level, and carbon dioxide (CO_2) can also trigger dosing pumps, chillers, valves, cooling devices, and circulation equipment. A faulty measurement is therefore not only a data-quality issue; it can become a physical command that perturbs the nutrient solution or growth environment.

Standard filtering and anomaly-detection methods usually score the signal. Closed-loop operation also requires a second question: whether the current value is trustworthy enough to update the controller state or authorize an actuator. A single high-pH spike should not authorize acid dosing, and a temporary EC excursion should not be treated the same as a persistent concentration shift. This motivates actuation-aware sensor validation, where logging, state update, rectification, and actuation authorization are separate decisions.

We present a closed-loop NFT hydroponic platform and propose AASVR-R, an actuation- and response-aware sensor-validation layer, as its fault-resilience mechanism. The contribution is structural: AASVR-R formalizes a supervisory measurement-to-actuation interface in which robust gates decide whether a sample is plausible, a state machine decides whether it can update the trusted state, a rectifier decides what value is exposed to control, an authorization rule decides whether actuation is allowed, and a response residual checks whether the measured process reacts as expected after an authorized command. The contributions are:

1. an integrated hydroponic deployment dataset and sensor-to-actuator formulation for NFT control;
2. AASVR-R, a streaming validation, rectification, authorization, and actuator-response verification method with explicit trust scoring and bounded updates;
3. a held-out synthetic-fault protocol with validation-only tuning, tuned baselines, ablations, component analysis, and paired statistical tests;
4. a reproducible Docker and LaTeX workflow that regenerates the tables, figures, and diagnostic outputs.

2. Related Work

2.1. Sensor faults in closed-loop systems

Sensor data-quality reviews distinguish missingness, outliers, drift, stuck values, saturation, and inconsistent rates as recurring faults in deployed systems [1]. Agricultural IoT systems face the same problem when sensor failure or miscalibration propagates into monitoring or control decisions [2, 3]. In closed-loop hydroponics, these faults have downstream consequences because an erroneous measurement can change the actuator command. This motivates treating validation, rectification, and actuation as separate decisions rather than as one smoothing step.

2.2. Filtering and monitoring baselines

The baseline family used in this paper covers common online filters, process-monitoring methods, model-based statistical tests, and lightweight one-class anomaly detectors. Hampel-style robust outlier detection uses robust scale estimates to suppress isolated deviations [4]. Kalman filtering gives a probabilistic recursive estimator for linear dynamical systems [5]. EWMA and CUSUM-style monitoring are standard approaches for persistent shifts [6, 7]. Generalized likelihood ratio (GLR) tests provide a classical statistical framework for abrupt change detection [8]. PCA and recursive PCA-style residual monitoring use residual statistics to capture coordinated changes or adaptive operating regimes [9, 10, 11]. Isolation Forest, One-Class SVM, and local outlier factor add common one-class machine-learning comparisons for anomaly detection [12, 13, 14]. Existing hydroponic automation studies have demonstrated reliable pH/EC regulation and IoT monitoring, but they generally treat validation as part of measurement filtering or rule execution rather than as a separate authorization decision [15, 16]. AASVR is complementary to these methods because it adds a control-facing authorization interface.

2.3. External process datasets

Public process datasets are useful stress tests for measurement validation, but their labels do not necessarily represent hydroponic sensor faults. HAI 21.03 contains industrial-control attack data from a hardware-in-the-loop testbed [17], and SKAB contains multivariate process-loop experiments with anomaly and changepoint labels [18]. These datasets are retained in the reproducibility artifacts as supplementary transfer checks, not as headline evidence. SWaT, WaDi, and Tennessee Eastman remain relevant larger benchmarks for future validation under their respective access and redistribution constraints [19, 20, 21].

3. Hydroponic System Design and Deployment

The target platform is a closed NFT hydroponic system for lettuce growth. NFT systems recirculate a shallow nutrient stream through sloped growth channels, allowing water and nutrient reuse when solution management is reliable [22, 23]. The deployed system is organized into two physical modules. The Growth Module houses the plants, lighting, air-temperature and humidity sensing, CO2 sensing, visual monitoring, and environmental actuation. The Nutrient Solution Container Module stores and recirculates the nutrient solution and contains the water-quality sensors and nutrient-solution actuators.

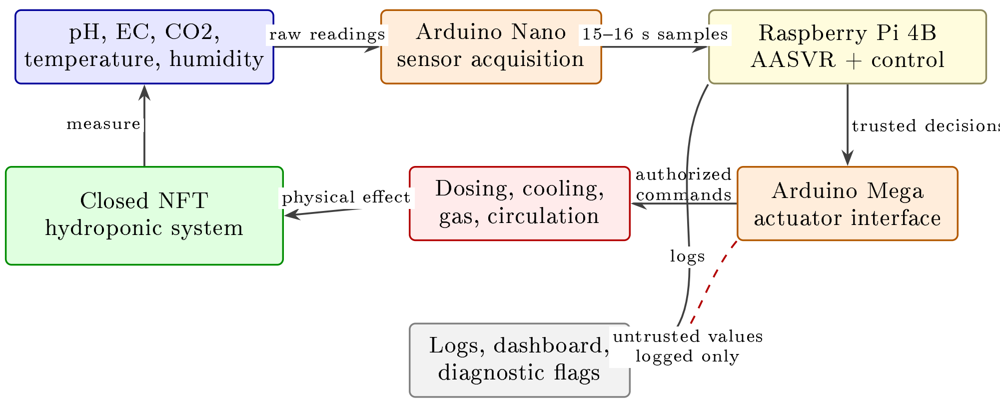


Figure 1: Hydroponic system architecture and AASVR location in the sensing-to-actuation path.

The Growth Module uses a DHT22 sensor for air temperature and humidity and an MH-Z19C nondispersive infrared CO2 sensor for gas concentration. CO2 regulation is implemented through a normally closed pneumatic solenoid valve connected to a CO2 cylinder. Air-temperature regulation is supported by a thermoelectric cooler. Two 40 W LED grow-light arrays provide a controlled light cycle, and a camera provides visual monitoring of the crop during the growth cycle.

The Nutrient Solution Container Module uses an Atlas Scientific analog pH kit, an Atlas Scientific Mini Conductivity K 1.0 kit for EC, a water-temperature sensor, and a JSN-SR04T waterproof ultrasonic distance sensor for water level. pH correction is implemented through acid/base dosing, and EC correction through concentrated nutrient solution or distilled-water addition. After a pH or EC dosing action, the dosing subsystem is disabled

Figure 2: Physical hydroponic system during operation.

for a 10-minute mixing interval before further dosing is allowed. A refrigerative water chiller regulates nutrient-solution temperature, an aerator runs throughout the growth cycle to support dissolved oxygen, and a water pump recirculates nutrient solution through the NFT channels.

The control unit connects sensors to an Arduino Nano and actuators to an Arduino Mega. Both microcontrollers communicate with a Raspberry Pi 4 Model B, which serves as the central control computer. This sensor-microcontroller-edge-computer structure is consistent with prior hydroponic automation systems, but the present paper focuses on the measurement-to-actuation reliability layer rather than on dashboarding as a contribution [15, 16]. The Raspberry Pi receives median-filtered sensor readings from the Arduino Nano, logs the readings, updates the ThingSpeak-backed dashboard, applies the validation/control logic, and sends authorized actuation commands to the Arduino Mega. The dashboard and camera feed are retained as deployment infrastructure; they are not claimed as the research contribution of this paper.

Table 1: Hydroponic platform components relevant to measurement validation and actuation. Water level is listed because it is part of the deployed system, but it is excluded from the primary quantitative analysis because the available raw values are not defensibly calibrated.

Subsystem	Measurements	Actuation or use
Growth module	Air temperature, humidity	Thermoelectric cooling
Growth module	CO2 concentration	CO2 solenoid valve
Growth module	Camera feed	Visual crop monitoring
Growth module	Light schedule	LED grow lights
Solution module	pH	Acid/base dosing, 10 min lockout
Solution module	EC	Nutrient/water dosing, 10 min lockout
Solution module	Water temperature	Refrigerative chiller
Solution module	Water level	Warning only in this analysis
Solution module	Recirculation state	Pump and aerator operation

The evaluation below treats this system as a deployed measurement-to-actuation stream: raw measurements are replayed through validation methods, transformed into trusted states, and then passed through an authorization rule before any actuator command is counted.

4. Problem Formulation

For hydroponic sensor s at time t, let $y_s(t)$ be the raw measurement and $\hat{x}_s(t)$ be the trusted value exposed to control. The method observes recent history $Y_s(t-w:t)$, broad physical limits $[P_{s,\min}, P_{s,\max}]$, a control band $[L_s^a, U_s^a]$, a rate limit r_s, and any available actuator state $a(t)$. In this system, s may correspond to pH, EC, humidity, air temperature, water temperature, or CO2, while $a(t)$ may represent dosing, cooling, circulation, or gas-release activity.

Table 2: Notation used in the AASVR formulation. Variables are defined for one sensor stream and then applied independently or jointly across the hydroponic sensor set.

Symbol	Meaning
$y_s(t)$	Raw value from sensor s at sample t
$\hat{x}_s(t)$	Trusted value exposed to the controller
$Y_s(t-w:t)$	Rolling window of recent raw values
$[P_{s,\min}, P_{s,\max}]$	Broad physically plausible range
$[L_s^a, U_s^a]$	Actuation or control band
r_s	Maximum plausible rate of change
u_s	Sensor uncertainty or minimum measurement resolution term
$q_s(t)$	Trust score in $[0,1]$
$z_s(t)$	Sensor state in the AASVR state set
$h_s(t)$	Evidence from the recent persistence/confirmation window
$A_s(t)$	Binary actuation authorization flag
θ	Tunable AASVR parameter vector

We define a measurement as *untrusted* when it is missing, physically implausible, rate-inconsistent, inconsistent with recent trusted state, or inconsistent with expected actuator effects. We define a *false actuation* as an authorization caused by untrusted evidence. A *missed actuation* is a failure to authorize action when the trusted state remains outside the control band.

The validation decision is separated from the control decision. Let $g_s(t) \in \{0,1\}$ denote whether a raw sample passes all measurement gates, and let $z_s(t)$ denote one of five states: normal, suspect transient, persistent deviation, accepted regime shift, and fault alert. The controller receives $\hat{x}_s(t)$ and $A_s(t)$, not the raw value alone. This separation is necessary because a value can be physically possible but still too uncertain to authorize actuation.

The conceptual evaluation objective is

$$\begin{aligned} J(\theta) = & w_1 E_{\text{pass}} + w_2 E_{\text{reject}} + w_3 A_{\text{false}} + w_4 A_{\text{missed}} \\ & + w_5 D_{\text{delay}} + w_6 T_{\text{unsafe}} + w_7 C_{\text{compute}}, \end{aligned} \tag{1}$$

where the terms respectively measure anomalous pass-through, false rejection, false actuation, missed actuation, detection delay, residence outside safe bands, and computational cost. The validation sweep uses the operative scalar

$$J_{\text{val}}(\theta) = \text{BA}(\theta) - 0.01 A_{\text{false}}(\theta) - 0.0005 F_{\text{alarm}}(\theta) - 0.001 D_{\text{delay}}(\theta), \tag{2}$$

where BA is balanced accuracy, A_{false} is the mean false-authorized actuation count, F_{alarm} is the mean false-alarm event count, and D_{delay} is mean detection delay in samples. Missed actuation, unsafe-band residence, and runtime are logged diagnostics rather than validation-selection terms in this study.

5. Proposed Method: AASVR-R

5.1. Overview

Figure 3 summarizes the AASVR-R pipeline. The method separates five decisions that are often conflated: measurement validation, trusted-state update, rectification, actuation authorization, and response verification. This separation is the core design choice. In the hydroponic system, this means that a suspicious pH value can be stored with a diagnostic flag while acid/base dosing remains suppressed until the value is persistent and trusted, and repeated dosing authority can be removed if the pH stream does not respond after a command.

5.2. Robust plausibility gate

For each sensor, AASVR-R estimates a robust scale over adjacent changes,

$$\sigma_{\Delta,s}(t) = 1.4826\,\text{MAD}\left(\Delta y_s(t-w:t)\right). \tag{3}$$

The accepted deviation tolerance is

$$\xi_s(t) = \max(\xi_{s,\min}, c_s \sigma_{\Delta,s}(t), r_s \Delta t, u_s). \tag{4}$$

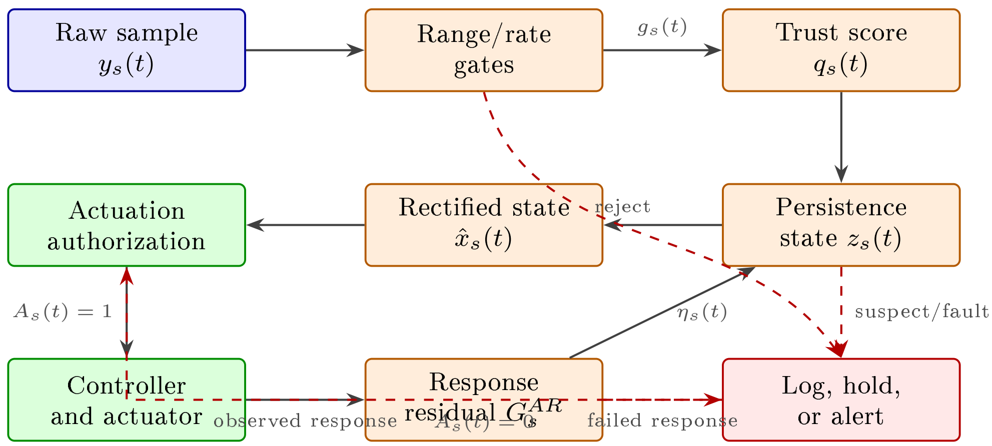


Figure 3: AASVR-R architecture. Measurement validation, trusted-state update, rectification, actuation authorization, and response verification are separate decisions.

A reading must satisfy broad physical range, maximum rate, and trusted-state deviation gates. The three elementary gates are

$$\begin{aligned} G_s^P(t) &= \mathbf{1}\{P_{s,\min} \le y_s(t) \le P_{s,\max}\}, \\ G_s^R(t) &= \mathbf{1}\{|y_s(t) - y_s(t-1)|/\Delta t \le r_s\}, \\ G_s^D(t) &= \mathbf{1}\{|y_s(t) - \hat{x}_s(t-1)| \le \xi_s(t)\}. \end{aligned} \tag{5}$$

A separate trend guard $G_s^T(t)$ rejects slow monotonic movement when no relevant actuator activity explains the direction of change. A stuck-value guard $G_s^S(t)$ rejects sustained exact repetition after a preceding change, which targets frozen plausible readings without treating ordinary low-variance noise as a fault. The overall gate is therefore

$$g_s(t) = G_s^P(t)G_s^R(t)G_s^D(t)G_s^T(t)G_s^S(t)G_s^M(t), \tag{6}$$

where $G_s^M(t)$ is one only when the value is present and parseable. The broad physical range is deliberately different from the crop-control band: for example, a pH value can be undesirable for lettuce while still being physically possible and therefore not automatically a sensor fault.

5.3. Trust score

AASVR-R also computes a continuous trust score so that control can distinguish weakly plausible values from strongly confirmed values. The implementation uses

$$q_s(t) = 0.25q_s^P(t) + 0.20q_s^R(t) + 0.20q_s^H(t) + 0.20q_s^A(t) + 0.15q_s^M(t), \quad (7)$$

where q_s^P is the physical-range score, q_s^R is the rate score, $q_s^H = \max(0, 1 - 0.2n_s^{\text{fail}})$ is the persistence score, q_s^A is actuator-consistency evidence, and q_s^M is the missingness score. Each component lies in $[0, 1]$. Trusted-deviation, uncommanded-trend, stuck-value, and actuator-response-residual failures cap the score through the gate outcome rather than adding another free weight. The deployed hydroponic configuration uses $q_{\min} = 0.70$, $c_s = 6.0$, transient limit 2, persistent limit 5, $k_s = 3$ confirming samples, and a four-sample replay cooldown for all primary sensors. The response-memory update uses $\lambda = 0.50$, with reliability thresholds 0.75 for high-risk dosing, 0.65 for medium-risk environmental actions, and 0.50 for low-risk actions. The physical pH/EC dosing code also contains a 10-minute mixing lockout.

5.4. State-machine persistence logic

Figure 4 shows the five persistence states. A single failed gate moves a sensor into a suspect transient state. Repeated failures create a persistent-deviation state. Consistent evidence can be accepted as a new process regime, while missing, stuck, saturated, or contradictory readings escalate to fault alert. This matters for hydroponics because real mixing and thermal responses can take multiple samples after dosing or cooling, whereas isolated spikes should not immediately change the trusted state.

The state update can be written as

$$z_s(t) = \mathcal{T}\left(z_s(t-1), g_s(t), q_s(t), h_s(t), a(t)\right), \quad (8)$$

where $h_s(t)$ summarizes persistence-window evidence. This representation makes the state machine explicit without treating every transition as a separate hand-tuned rule in the manuscript. The transition function $\mathcal{T}$ is implemented by Algorithm 1.

5.5. Rectification and actuation authorization

When a reading is accepted, AASVR-R sets $\hat{x}_s(t) = y_s(t)$. When it is suspect, the method either holds $\hat{x}_s(t-1)$ or applies a bounded update

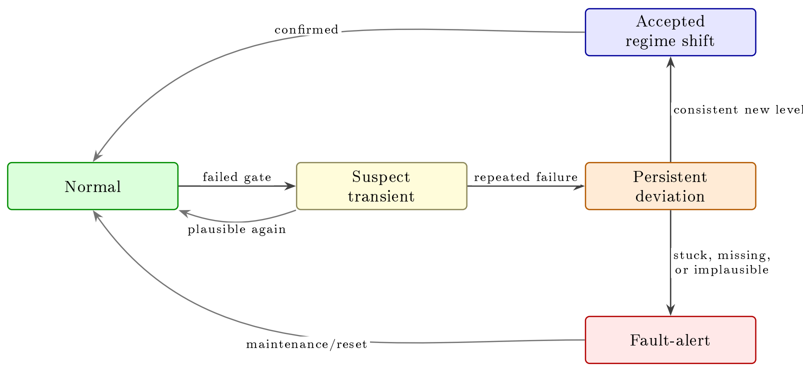


Figure 4: AASVR persistence state machine.

limited by $r_s \Delta t$:

$$\hat{x}_s(t) = \begin{cases} y_s(t), & z_s(t) \in \{\text{normal}, \text{accepted}\}, \\ \hat{x}_s(t-1), & z_s(t) = \text{suspect}, \\ \hat{x}_s(t-1) + \text{clip}(\Delta y_s, -r_s\Delta t, r_s\Delta t), & \text{bounded}, \\ \text{alert}, & z_s(t) = \text{fault}. \end{cases} \tag{9}$$

Actuation is authorized only if the trusted value is outside the control band, the violation persists for the configured number of samples, the sensor state is normal or accepted-regime-shift, the trust score is sufficient, the actuator is not in cooldown, and the recent response reliability $\eta_s(t)$ is above the risk-specific threshold:

$$\begin{aligned} A_s(t) = &\mathbf{1}\{\hat{x}_s(t) \notin [L_s^a, U_s^a]\} \mathbf{1}\{p_s(t) \geq k_s\} \mathbf{1}\{q_s(t) \geq q_{\min}\} \\ &\times \mathbf{1}\{\eta_s(t) \geq \eta_{\min,\rho(s)}\} \mathbf{1}\{z_s(t) \in \mathcal{Z}_{\text{safe}}\} \mathbf{1}\{\ell_s(t) = 0\}. \end{aligned} \tag{10}$$

Here $p_s(t)$ is the number of consecutive trusted control-band violations, k_s is the persistence requirement, $\rho(s)$ is the action-risk class, $\mathcal{Z}_{\text{safe}} = \{\text{normal}, \text{accepted}\}$, and $\ell_s(t)$ is the actuator lockout indicator. For pH and EC, this rule complements the 10-minute dosing lockout already present in the deployed system; for CO2 and temperature, it prevents transient readings from becoming immediate gas-release or cooling commands.

5.6. Actuator-response residual

After an authorization at time t_0, AASVR-R checks whether the trusted measurement moves in the expected direction within a response window. Let

$d_s(t_0) \in \{-1, 0, +1\}$ denote the expected direction: high pH or EC implies a decreasing response after acid or dilution, low pH or EC implies an increasing response after base or nutrient dosing, and cooling implies decreasing temperature. The response evidence is

$$M_s(t_0) = \max_{\tau_{\min} \le \tau \le \tau_{\max}} d_s(t_0) \left[\hat{x}_s(t_0 + \tau) - \hat{x}_s(t_0)\right]. \tag{11}$$

The response gate is

$$G_s^{AR}(t_0) = \mathbf{1}\{M_s(t_0) \ge \rho_s\}. \tag{12}$$

If this gate fails, the method logs an actuator-response residual, withholds further authority from the sensor according to the reliability memory, and may escalate to fault alert. The reliability memory is

$$\eta_s(t+1) = \lambda \eta_s(t) + (1 - \lambda) c_s(t), \tag{13}$$

where $c_s(t) = 1$ after a confirmed response and $c_s(t) = 0$ after a failed response. This term is intentionally different from $q_s(t)$: $q_s(t)$ scores the current sample, while $\eta_s(t)$ scores whether recent actuation episodes produced credible sensor response.

Informal proposition 1. Assume the true process state satisfies $|x_s(t) - x_s(t-1)| \le r_s \Delta t$ and the previous trusted estimate has error $|\hat{x}_s(t-1) - x_s(t-1)| \le e_{t-1}$. Under the bounded rectification branch in Eq. (9), $|\hat{x}_s(t) - x_s(t)| \le e_{t-1} + 2 r_s \Delta t$ for any raw value $y_s(t)$, including arbitrarily large excursions. The result follows from applying a clipped update of magnitude at most $r_s \Delta t$ while the true state also moves by at most $r_s \Delta t$. Under a sustained suspect state, this per-sample bound can accumulate linearly, which is the formal counterpart of the empirical drift weakness. This is therefore a bounded-input statement for the controller-facing estimate, not a full closed-loop stability proof.

Algorithm 1 Streaming AASVR-R update for one sensor

Require: Raw value $y_s(t)$, previous trusted state $\hat{x}_s(t-1)$, recent history, limits, actuator state.

Estimate robust local scale $\sigma_{\Delta,s}(t)$ and tolerance $\xi_s(t)$.
Apply physical-range, rate, trusted-deviation, missingness, trend, and stuck-value gates.
If actuator response is pending, check the response-window residual and update $\eta_s(t)$.
Update the persistence state from the gate outcome.
if the reading is accepted **then**
    Set $\hat{x}_s(t) \leftarrow y_s(t)$.
else if the reading is transient suspect **then**
    Hold or bounded-predict $\hat{x}_s(t)$.
else
    Keep the last safe estimate and log a fault-alert decision.
end if
Authorize actuation only if trusted state, persistence, score, reliability, and cooldown rules are satisfied.
return Trusted value, sensor state, trust score, actuation flag, and log record.

6. Evaluation Protocol

6.1. Hydroponic deployment data

The hydroponic data contain 121244 raw rows collected from 2024-02-27 to 2024-03-28 at a median cadence of approximately 16 seconds. The primary analysis uses EC, pH, humidity, air temperature, water temperature, and CO2. Water level is part of the deployed system but is excluded from the primary analysis because the available raw values are not defensibly calibrated.

6.2. Supplementary external stress tests

HAI 21.03 and SKAB are retained in the reproducibility outputs as supplementary stress tests rather than as main evidence. Their labels describe cyber-physical attacks and process-loop anomalies, not hydroponic sensor faults. The direct-transfer results are therefore reported only as a limitation check.

6.3. Fault injection protocol

Synthetic faults are injected into windows sampled from the real hydroponic background. For each sensor, fault type, magnitude, and duration, three deterministic replicates are generated and assigned to tune, validation, and test splits. The reported hydroponic benchmark uses only the held-out test split; AASVR-R and baseline hyperparameters are selected on validation trials. Fault windows include 90 pre-fault samples and 120 post-fault samples so that detection delay and false alarms are evaluated around the injected event rather than in isolation.

Table 3: Synthetic fault-injection protocol for the hydroponic replay benchmark. For each protocol row, durations are 1, 5, and 30 samples with one replicate assigned to each split. Magnitudes are in sensor units: pH uses 0.5/2.0, CO2 500/2000 ppm, EC 100/400, air temperature 2/8 °C, and water temperature 1/4 °C.

Fault type	Injection rule over fault window $\mathcal{I}$	Purpose
Spike	$y'_s(t) = y_s(t) + m$	Single-window offset
Multi-spike	$y'_s(t) = y_s(t) + b_t m,\ b_t \in \{-1, 1\}$	Alternating impulse faults
Dropout	$y'_s(t) = \mathrm{NaN}$	Missing samples
Drift	$y'_s(t) = y_s(t) + m(t - t_0)/\lvert\mathcal{I}\rvert$	Slow monotonic deviation
Bias	$y'_s(t) = y_s(t) + m$	Persistent offset
Noise burst	$y'_s(t) = y_s(t) + \epsilon_t,\ \epsilon_t \sim \mathcal{N}(0, m^2)$	Increased variance
Step	$y'_s(t) = y_s(t) + m$	Abrupt sustained change
Stuck-at	$y'_s(t) = y_s(t_0)$	Constant repeated value
Saturation	$y'_s(t) = m$	Sensor rail or implausible fixed value

6.4. Counterfactual response replay

A second replay evaluates the actuator-response residual. Replay-derived controller authorizations are treated as pseudo-actuator commands because the available feed does not contain real actuator-state columns. Healthy trials inject a first-order response in the expected direction after authorization. Response-fault trials inject no-response, stuck-after-command, weak-response, delayed-response, and wrong-direction behavior. These trials test whether the supervisor detects missing measurement response and suppresses repeated authority; they are not interpreted as measured physical closed-loop disturbances.

6.5. Baselines and metrics

The baselines are raw thresholding, moving average, moving median, Hampel, Kalman, EWMA, CUSUM, PCA-style monitoring, GLR, recursive

PCA, Isolation Forest, One-Class SVM, and local outlier factor. A duplicated physical-threshold rule remains in the reproducibility outputs but is not interpreted as a separate scientific baseline. Every displayed baseline is tuned on the validation split using Eq. (2) and then wrapped in the same persistence/cooldown authorization supervisor for false-actuation scoring. The hydroponic synthetic-fault benchmark reports precision, recall, specificity, balanced accuracy, detection delay, false alarm events, false actuation mean, and false actuation standard deviation.

Bootstrap confidence intervals use 1000 resamples of injected-fault trials, not individual samples. Paired Wilcoxon signed-rank tests compare AASVR-R against each baseline on per-trial balanced accuracy and false actuation, with Holm correction applied separately across the 13-baseline balanced-accuracy family and the 13-baseline false-actuation family. Rank-biserial correlation and paired d_z are reported in the reproducibility tables, and a sign test is also reported for false-actuation differences because that distribution is zero-inflated and skewed. Sensor–fault grids use Benjamini–Hochberg correction.

Table 4: Baseline taxonomy used in the hydroponic benchmark. The final column marks whether a method explicitly separates anomaly detection from actuation authorization.

Family	Methods	Primary role	Act.-aware
Rule	Raw threshold	Existing physical-limit control logic	No
Filters	Moving average, moving median, Hampel, Kalman	Noise and spike suppression	No
Process monitoring	EWMA, CUSUM, PCA, GLR, recursive PCA	Shift or process-deviation detection	No
One-class ML	Isolation Forest, One-Class SVM, LOF	Data-driven anomaly boundary	No
Proposed method	AASVR-R	Validation, rectification, authorization, and response verification	Yes

7. Results

The results are organized as follows. Section 7.1 characterizes the hydroponic streams. Section 7.2 reports the held-out synthetic-fault benchmark

Table 5: Hydroponic measurement streams used in the primary AASVR analysis. The table separates deployed sensing from quantitative use so that the water-level exclusion is explicit rather than hidden.

Variable	Module	Primary analysis	Control relevance
EC	Solution	Yes	Nutrient/water dosing evidence
pH	Solution	Yes	Acid/base dosing evidence
Humidity	Growth	Yes	Growth-environment monitoring
Air temperature	Growth	Yes	Air-cooling evidence
Water temperature	Solution	Yes	Chiller evidence
CO2	Growth	Yes	CO2 valve evidence
Water level	Solution	No	Warning channel; calibration unresolved

and paired tests. Section 7.3 breaks performance down by sensor and fault class. Section 7.4 separates the contribution of gating, rectification, and authorization. Section 7.5 reports ablations and tuning sensitivity, and Section 7.6 connects the aggregate benchmark to representative deployment traces.

7.1. Hydroponic deployment characterization

The deployment data are first used to characterize the real measurement background on which AASVR operates. The 121244-row feed covers a continuous 30-day lettuce growth period, with approximately 16-second median sampling. The key point for this paper is not that the dashboard collected data, but that the closed-loop system generated the kind of sensor-to-actuator stream in which a bad reading can trigger a physical correction.

7.2. Hydroponic synthetic-fault benchmark

Table 6 summarizes the primary target-domain result on 270 held-out synthetic-fault trials. LOF is the strongest point detector and is fractionally higher than AASVR-R on balanced accuracy. AASVR-R has lower recall than LOF but higher specificity and far fewer alert samples, reflecting a more conservative controller-facing validator. Relative to the base AASVR supervisor, AASVR-R preserves nearly the same balanced accuracy while reducing mean false-authorized actuation through response-memory gating. One-Class SVM also has low false-authorized actuation, but it obtains this by rejecting aggressively, with a false-positive rate of 0.667 and nearly 890 alert samples per trial. The contribution claim is therefore not that AASVR-R dominates every point-detection metric; it is that the method exposes validation, rectification, authorization, and response verification as auditable controller-facing decisions.

Table 6: Held-out hydroponic synthetic-fault benchmark. Baselines are validation-tuned and evaluated on the same test split with a shared persistence/cooldown supervisor. Bold entries mark the best value in each metric column among the displayed methods; lower is better for false actuation.

Method	Recall	Specificity	Bal. acc.	FPR	Delay	False act.	False act. SD
LOF	0.853	0.798	**0.826**	0.202	0.270	0.130	0.629
AASVR	0.776	0.873	0.824	0.127	**0.000**	0.259	0.853
AASVR-R	0.776	0.870	0.823	0.130	**0.000**	**0.044**	**0.295**
Kalman	0.613	0.803	0.708	0.197	**0.000**	0.515	1.175
Isolation Forest	0.581	0.826	0.703	0.174	0.126	0.630	1.277
Hampel	0.411	0.984	0.698	0.016	**0.000**	1.041	1.502
Moving median	0.411	0.984	0.698	0.016	**0.000**	0.993	1.496
Recursive PCA	0.317	0.988	0.652	0.012	**0.000**	1.319	1.653
One-Class SVM	**0.912**	0.333	0.623	0.667	0.089	0.070	0.403
GLR	0.129	**0.999**	0.564	**0.001**	**0.000**	1.330	1.638

Table 7: Paired held-out effects for AASVR-R against representative methods. Positive ΔBA means AASVR-R has higher per-trial balanced accuracy; positive Δfalse actuation means the comparison method has more false actuations than AASVR-R. d_z is the paired standardized BA effect size; $p_{\mathrm{FA,sign}}$ is a one-sided sign test for the zero-inflated false-actuation difference.

Baseline	ΔBA	d_z	p_{BA}	ΔFalse act.	95% CI	$p_{\mathrm{FA,sign}}$
Base AASVR	-0.001	-0.722	1.000	0.215	[0.126, 0.322]	< 0.001
LOF	-0.003	-0.015	0.020	0.085	[0.026, 0.152]	0.025
One-Class SVM	0.200	1.011	< 0.001	0.026	[-0.030, 0.081]	0.304
Kalman	0.115	0.453	< 0.001	0.470	[0.337, 0.604]	< 0.001
Hampel	0.125	0.490	< 0.001	0.996	[0.818, 1.181]	< 0.001
CUSUM	0.183	0.767	< 0.001	1.230	[1.044, 1.422]	< 0.001

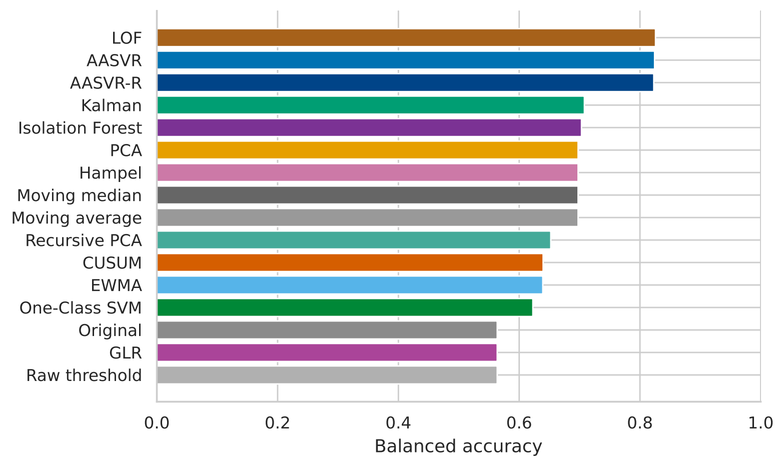


Figure 5: Held-out hydroponic synthetic-fault balanced accuracy across methods. LOF is the strongest point detector; AASVR-R is close while using an interpretable response-aware supervisory structure.

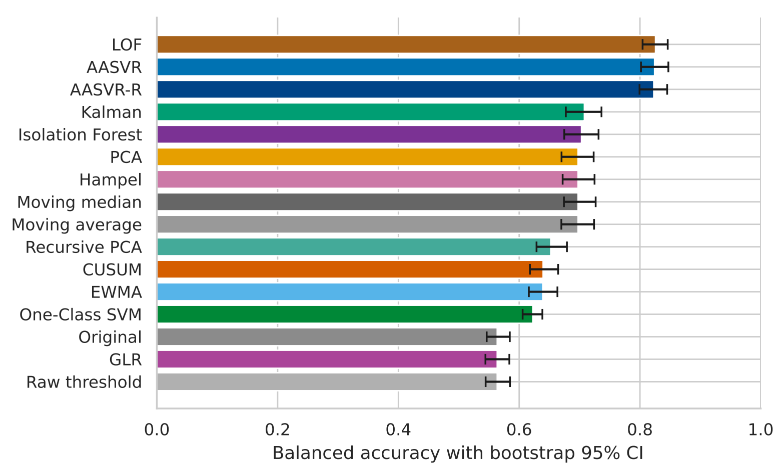


Figure 6: Bootstrap uncertainty for balanced accuracy. Intervals are trial-level resamples.

7.3. Sensor-wise and fault-type comparisons

Table 8 breaks down AASVR-R performance by hydroponic sensor. CO2, water temperature, and air temperature faults are detected with balanced accuracy above 0.84. EC is the weakest stream, mainly because gradual concentration faults are hard to distinguish from plausible nutrient-solution changes without independent reference instrumentation. Response-memory gating reduces pH/EC false authorization in the replay, but it does not solve the EC detection problem.

Table 9 and Figure 7 show performance by injected fault class. AASVR-R performs best on saturation, dropout, bias, spike, noise burst, step, and multi-spike faults. Drift and stuck-at faults remain difficult because they can resemble slow process movement or repeated steady values without an independent reference. The response-memory extension reduces the stuck-at false-authorization tail relative to the base supervisor, but stuck-at remains the weakest detection case.

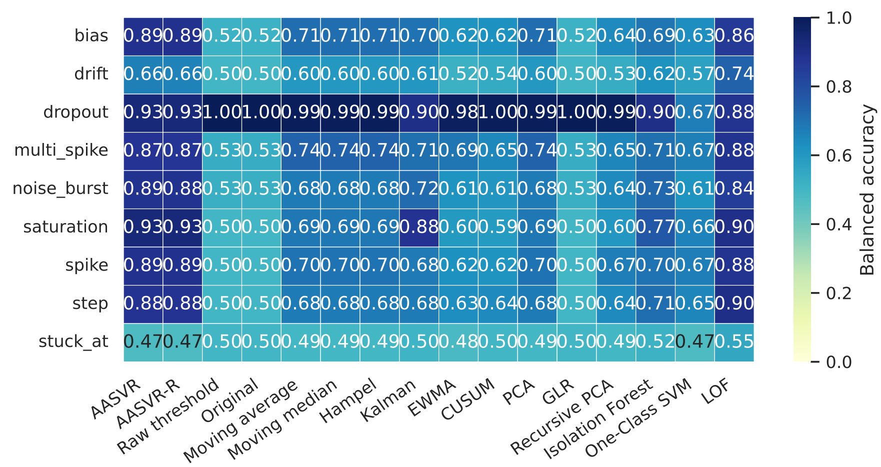


Figure 7: Hydroponic synthetic-fault heatmap by sensor and fault type. Slow drift and stuck-at behavior remain limitations.

Table 8: AASVR-R synthetic-fault performance by hydroponic sensor. False actuation is reported as mean events per injected-fault trial.

Sensor	Recall	Specificity	Bal. acc.	False act.
CO2	**0.848**	**0.882**	**0.865**	0.056
Water temperature	0.820	0.871	0.846	0.037
Air temperature	0.822	0.862	0.842	0.093
pH	0.781	0.868	0.825	**0.019**
EC	0.609	0.866	0.737	**0.019**

Table 9: AASVR-R synthetic-fault performance by injected fault type. Stuck-at and drift are the most difficult conditions because frozen plausible values and slow monotonic movement can resemble real process behavior.

Fault type	Recall	Specificity	Bal. acc.	False act.
Saturation	**1.000**	0.864	**0.932**	**0.000**
Dropout	**1.000**	0.860	0.930	**0.000**
Bias	0.910	0.868	0.889	**0.000**
Spike	0.896	**0.880**	0.888	0.033
Noise burst	0.893	0.875	0.884	0.033
Step	0.874	0.878	0.876	**0.000**
Multi-spike	0.872	0.868	0.870	**0.000**
Drift	0.464	0.862	0.663	0.067
Stuck-at	0.074	0.873	0.474	0.267

7.4. Component contribution and ablation

Table 10 separates the contribution of the supervisory components on the same 270 held-out test trials as Table 6. Raw thresholding with the shared supervisor has high specificity but poor detection because only physically extreme values are rejected. Adding robust gates and hold rectification raises balanced accuracy and reduces false-authorized actuation. The trust-score-only row is identical to robust gates plus hold, showing that in the selected configuration the trust score is mainly diagnostic unless it is coupled to the full persistence and authorization logic. AASVR-R keeps the point-detection score close to the base supervisor while using response reliability to reduce the false-authorization tail.

Table 11 shows the internal ablation results on a deterministic 120-trial

Table 10: Component-contribution analysis on the full 270-trial held-out synthetic-fault test split. Values are mean events or scores per trial.

Variant	Recall	Specificity	Bal. acc.	False act.
Raw threshold + supervisor	0.129	**0.999**	0.564	1.330
Robust gates + hold	0.737	0.887	0.812	0.363
Robust gates + trust + hold	0.737	0.887	0.812	0.363
Full AASVR	0.776	0.873	**0.824**	0.259
Full AASVR-R	0.776	0.870	0.823	**0.044**
AASVR without cooldown	0.776	0.873	**0.824**	0.541
AASVR without confirmation/cooldown	0.776	0.873	**0.824**	2.207

Table 11: AASVR ablation summary on sampled held-out test trials. Rows are ordered to emphasize false-actuation impact relative to the full method. Detection scores can remain similar while control outcomes change, which supports evaluating control-aware metrics rather than point-wise detection alone.

Variant	Bal. acc.	Recall	False act.
No actuation confirmation	**0.820**	0.765	0.583
Bounded prediction mode	0.774	0.669	0.575
No cooldown	**0.820**	0.765	0.475
No trend guard	0.804	0.728	0.325
Full AASVR	**0.820**	0.765	0.233
No stuck detector	0.819	0.757	0.233
No robust MAD scale	0.819	**0.794**	0.192

subset of the held-out test split. It is not numerically identical to the full 270-trial reference in Tables 6 and 10, and the caption states the subset explicitly to avoid interpreting small differences as algorithm changes. Several ablations preserve similar detection scores but change control behavior. Removing actuation confirmation increases false actuation from 0.233 to 0.583 events per trial in that subset, and removing cooldown increases it to 0.475. The stuck-detector contrast is better interpreted through the stuck-at-specific result in Table 9; its aggregate effect is small because stuck-at is one of several injected fault classes.

The validation sweep in Table 12 shows that the selected AASVR configuration is not an isolated optimum. Multiple trust-score and transient-limit settings produced the same validation objective when the robust scale multiplier was 6.0. These parameters were selected before evaluating the held-out test split in Table 6. A transfer check over the top validation settings showed held-out balanced accuracy within 0.0011 across the top 15 settings on a fixed sampled test subset; objective-weight profiles that increased the false-actuation penalty selected the same top setting in the current grid. This indicates that the reported result is not sensitive to the arbitrary choice among tied $q_{\min}$ or transient-limit settings.

Table 12: Top AASVR parameter settings from the validation split. The selected setting uses $c_s = 6.0$, $q_{\min} = 0.70$, and transient limit 2; tied settings are retained to show sensitivity.

c_s	$q_{\min}$	Transient limit	Bal. acc.	Objective
6.0	0.70	2	**0.824**	**0.744**
6.0	0.70	1	**0.824**	**0.744**
6.0	0.55	3	**0.824**	**0.744**
6.0	0.55	2	**0.824**	**0.744**
6.0	0.55	1	**0.824**	**0.744**

7.5. Counterfactual response replay

Table 13 reports the actuator-response replay. Base AASVR has no mechanism to evaluate whether the measurement responded after a command, so it detects none of the injected response failures. AASVR-R detects no-response and stuck-after-command cases consistently, detects some delayed and weak responses, and preserves all healthy-response trials in this replay. Wrong-direction faults are usually suppressed by existing plausibility and reliability logic rather than explicitly labeled as response residuals, so they remain a target for stronger direction-specific deployment logs.

Table 13: Counterfactual actuator-response replay. Response correctness is the mean of fault detection on response-fault trials and no false residual on healthy trials. Authorizations are replay-derived controller commands, not measured physical actuator events.

Method	Response correctness	Fault alert rate	Authorizations	Final η_s
Base AASVR	0.167	0.000	4.233	1.000
AASVR-R	**0.700**	**0.533**	**1.633**	0.650

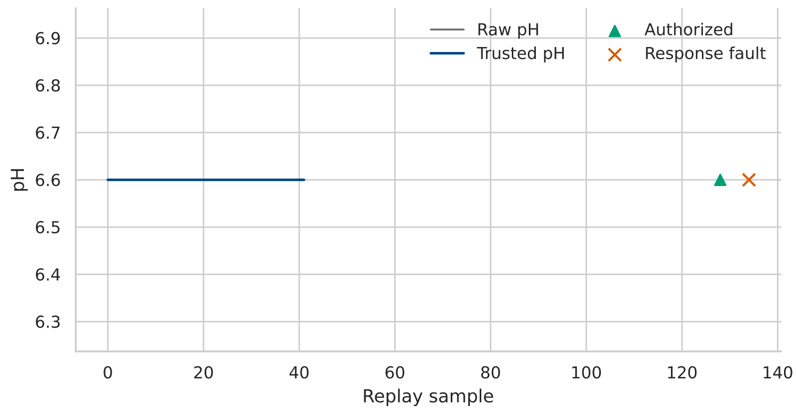
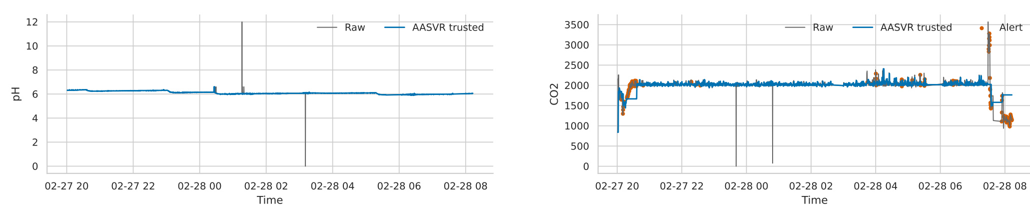


Figure 8: Counterfactual pH no-response replay. AASVR-R authorizes from a trusted high-pH violation, opens a response window, and flags a response residual when the trusted pH fails to decrease.

7.6. Representative hydroponic trace

Figure 9 shows replay segments from the hydroponic deployment. The traces are included to connect the aggregate benchmark to the actual deployment setting. They illustrate the measurement-to-state transformation that precedes any actuation decision.



(a) pH replay

(b) CO2 replay

Figure 9: Representative hydroponic replay traces. The raw streams and trusted estimates are plotted over early deployment segments. These traces are not used as ground-truth fault labels; they show how the method exposes a trusted state rather than passing every raw measurement directly to the controller.

Figure 10 and Table 14 show the audit trail for a held-out pH fault replay. The raw pH value jumps from the trusted state near 6.09 to a faulted value above 8.0. The physical and rate gates remain true, but the trusted-deviation gate fails. The supervisor holds the trusted value, lowers the trust score, transitions from suspect transient to persistent deviation and then fault alert, and never authorizes actuation from the untrusted pH evidence.

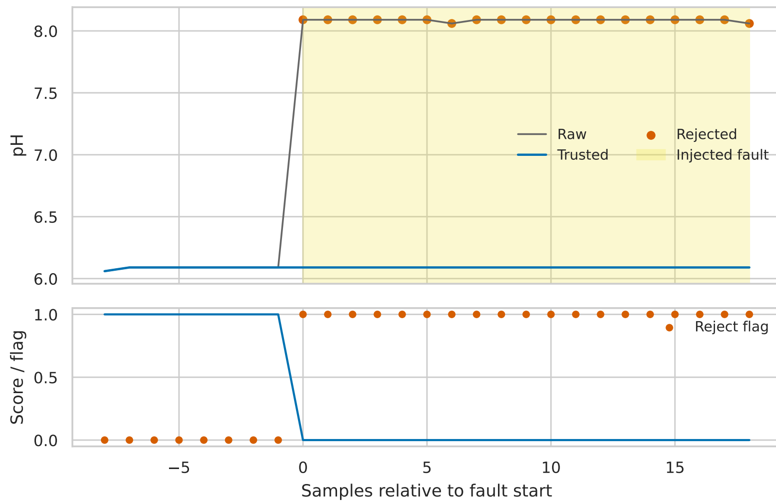


Figure 10: AASVR audit trail for a held-out pH fault replay. The shaded region is the injected fault window; rejected samples are logged without authorizing actuation.

Table 14: Audit-trail excerpt for the pH replay in Fig. 10. G^D is the trusted-deviation gate.

Rel. sample	Raw	Trusted	G^D	State	Action	Authorized
-1	6.09	6.09	True	normal	accept	False
0	8.09	6.09	False	suspect	hold	False
2	8.09	6.09	False	persistent	hold	False
4	8.09	6.09	False	fault	alert	False
8	8.09	6.09	False	fault	alert	False

8. Discussion

8.1. Why actuation awareness matters

The ablation, component, response-replay, and audit-trail results show why actuation-aware validation should be evaluated with control metrics and decision transparency. LOF remains the best point detector in this benchmark, and after applying the same cooldown supervisor it also has a low false-authorized actuation count. AASVR-R's contribution is different: it makes validation, trusted-state update, rectification, authorization, and post-command response verification explicit and auditable, and it does so without a learned one-class boundary. The trust-score-only component result is also informative. In this configuration it does not change the robust-gate baseline because the binary gates already determine most accept/reject decisions; its value appears mainly as a diagnostic and as an input to the full state/authorization supervisor. For hydroponic systems, these distinctions are practical because a logged anomaly, a held trusted state, a pump command, and a failed post-command response are different engineering decisions.

8.2. External transfer

The supplementary HAI and SKAB checks show that the same streaming interface can run on independent process datasets, but direct parameter transfer is weak. HAI attacks and SKAB process anomalies are not identical to hydroponic sensor faults, and the current method uses hydroponic physical ranges and rate assumptions. These results are therefore not used as evidence of cross-domain dominance; they identify the need for process-specific limits or model residuals before claiming industrial transfer.

8.3. Limitations

The hydroponic experiment does not provide independent reference instruments for every sensor at every time point. Synthetic fault injection is therefore necessary for controlled scoring, but injected faults cannot cover every real failure mode. The reported false-actuation values are replayed authorization events, not measured physical process disturbances, because the replay benchmark does not close the loop around real actuators. The response replay is also counterfactual: replay-derived authorizations stand in for actuator commands because the available feed does not contain real actuator-state columns. Slow drift, stuck-at behavior, weak response, and EC-bias cases remain the principal weaknesses because they can resemble real nutrient-solution dynamics without reference instrumentation or measured actuator effects. AASVR-R reduces repeated authority after failed response and substantially reduces the stuck-at false-authorization tail, but it does not make frozen plausible values fully detectable when no command-response episode occurs. Water level is excluded from primary analysis because the available raw values are not defensibly calibrated. Crop outcomes are treated only as deployment context because enclosure, lighting, cooling, nutrient delivery, carbon dioxide regulation, and automation are confounded.

9. Conclusion

We presented a closed-loop hydroponic control system and proposed AASVR-R as its response-aware sensor validation and rectification layer. The method separates validation, state update, rectification, actuation authorization, and response verification so that untrusted measurements can be logged without automatically driving physical commands. On held-out hydroponic synthetic faults, AASVR-R was competitive with the strongest tuned detector while offering a transparent supervisory structure and high-specificity controller-facing estimates. Counterfactual response replay showed that actuator-response residuals address no-response and stuck-after-command cases that single-stream gates miss. The remaining failures on slow drift, weak response, and plausible stuck values show where future deployments should add real actuator-state logs and reference measurements.

Data Availability

The repository contains code, configuration files, processed-data descriptors, Docker workflow files, and paper-generation assets. Raw hydroponic and external benchmark files are not committed to the repository. Public datasets can be obtained from their original providers, and request-access datasets should be obtained under the providers' terms.

Declaration of Competing Interest

The authors declare no competing interests for the current manuscript draft.

Declaration of Generative AI and AI-Assisted Technologies

AI-assisted tools were used to help organize code, draft text, and check consistency. The authors remain responsible for the scientific content, claims, results, and final manuscript.

References

[1] H. Y. Teh, A. W. Kempa-Liehr, K. I.-K. Wang, Sensor data quality: a systematic review, Journal of Big Data 7 (11) (2020). doi:10.1186/s40537-020-0285-1.

[2] X. Zou, W. Liu, Z. Huo, S. Wang, Z. Chen, C. Xin, Y. Bai, Z. Liang, Y. Gong, Y. Qian, L. Shu, Current status and prospects of research on sensor fault diagnosis of agricultural internet of things, Sensors 23 (5) (2023) 2528. doi:10.3390/s23052528.

[3] Y.-B. Lin, Y.-W. Lin, J.-Y. Lin, H.-N. Hung, Sensortalk: An iot device failure detection and calibration mechanism for smart farming, Sensors 19 (21) (2019) 4788. doi:10.3390/s19214788.

[4] F. R. Hampel, The influence curve and its role in robust estimation, Journal of the American Statistical Association 69 (346) (1974) 383–393. doi:10.1080/01621459.1974.10482962.

[5] R. E. Kalman, A new approach to linear filtering and prediction problems, Journal of Basic Engineering 82 (1) (1960) 35–45. doi:10.1115/1.3662552.

[6] S. W. Roberts, Control chart tests based on geometric moving averages, Technometrics 1 (3) (1959) 239–250. doi:10.1080/00401706.1959.10489860.

[7] E. S. Page, Continuous inspection schemes, Biometrika 41 (1/2) (1954) 100–115. doi:10.1093/biomet/41.1-2.100.

[8] M. Basseville, I. V. Nikiforov, Detection of Abrupt Changes: Theory and Application, Prentice Hall, 1993.

[9] H. Hotelling, Multivariate quality control, Techniques of Statistical Analysis (1947) 111–184.

[10] J. E. Jackson, G. S. Mudholkar, Control procedures for residuals associated with principal component analysis, Technometrics 21 (3) (1979) 341–349. doi:10.1080/00401706.1979.10489779.

[11] W. Li, H. H. Yue, S. Valle-Cervantes, S. J. Qin, Recursive pca for adaptive process monitoring, Journal of Process Control 10 (5) (2000) 471–486. doi:10.1016/S0959-1524(00)00022-6.

[12] F. T. Liu, K. M. Ting, Z.-H. Zhou, Isolation forest, in: 2008 Eighth IEEE International Conference on Data Mining, IEEE, 2008, pp. 413–422. doi:10.1109/ICDM.2008.17.

[13] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, R. C. Williamson, Estimating the support of a high-dimensional distribution, Neural Computation 13 (7) (2001) 1443–1471. doi:10.1162/089976601750264965.

[14] M. M. Breunig, H.-P. Kriegel, R. T. Ng, J. Sander, LOF: Identifying density-based local outliers, in: Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data, 2000, pp. 93–104. doi:10.1145/342009.335388.

[15] K. Tatas, A. Al-Zoubi, N. Christofides, C. Zannettis, M. Chrysostomou, S. Panteli, A. Antoniou, Reliable iot-based monitoring and control of hydroponic systems, Technologies 10 (1) (2022) 26. doi:10.3390/technologies10010026.

[16] D. S. Domingues, H. W. Takahashi, C. A. Camara, S. L. Nixdorf, Automated system developed to control ph and concentration of nutrient solution evaluated in hydroponic lettuce production, Computers and Electronics in Agriculture 84 (2012) 53–61. doi:10.1016/j.compag.2012.02.006.

[17] H.-K. Shin, W. Lee, J.-H. Yun, H. Kim, Hai 2021: A new dataset for anomaly detection in industrial control systems, in: Proceedings of the 14th Cyber Security Experimentation and Test Workshop, 2021, pp. 1–9. doi:10.1145/3474718.3474719.
URL `https://doi.org/10.1145/3474718.3474719`

[18] I. D. Katser, V. O. Kozitsin, Skoltech anomaly benchmark (skab), `https://www.kaggle.com/dsv/1693952` (2020). doi:10.34740/KAGGLE/DSV/1693952.

[19] Secure water treatment (swat) dataset, https://www.sutd.edu.sg/itrust/itrust-labs/datasets/dataset-characteristics/swat/, accessed 2026-05-18 (2026).

[20] Water distribution (wadi) dataset, https://www.sutd.edu.sg/itrust/itrust-labs/datasets/dataset-characteristics/wadi/, accessed 2026-05-18 (2026).

[21] Tennessee eastman reference data for fault-detection and decision support systems, https://data.dtu.dk/articles/dataset/Tennessee_Eastman_Reference_Data_for_Fault-Detection_and_Decision_Support_Systems/13385936, accessed 2026-05-18 (2026).

[22] G. Niu, J. Masabni, Hydroponics, in: Plant Factory Basics, Applications and Advances, Academic Press, 2022, pp. 153–166. doi:10.1016/B978-0-323-85152-7.00014-6.

[23] M. Asaduzzaman, G. Niu, T. Asao, Nutrients recycling in hydroponics: Opportunities and challenges toward sustainable crop production under controlled environment agriculture, Frontiers in Plant Science 13 (2022) 845472. doi:10.3389/fpls.2022.845472.